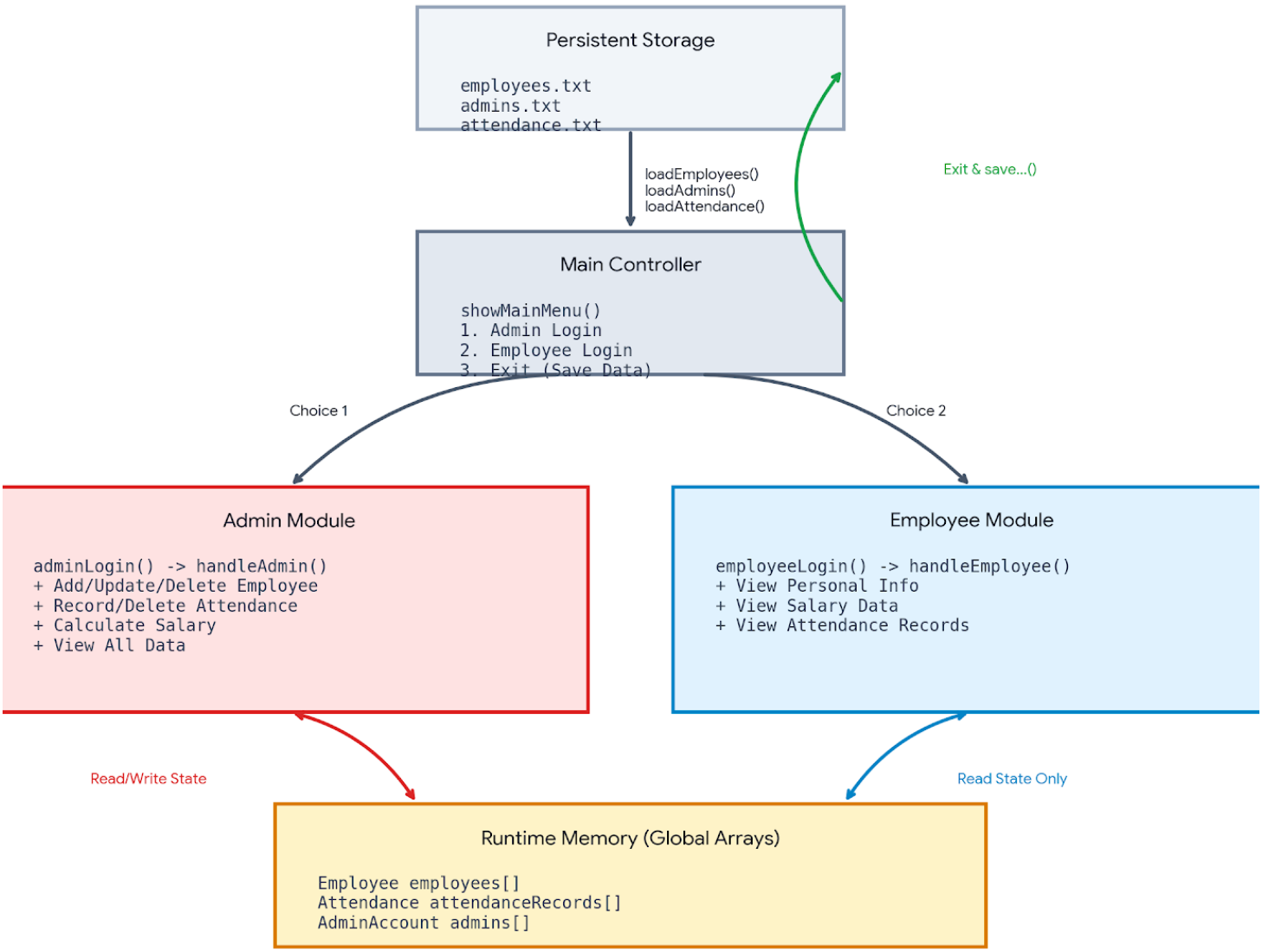
System Architecture & Control Flow: Employee Payroll Management



Employee Payroll Management System

1. Team Information

Full Name	Student ID	Program
Mahmoud Walid Mohamed Aziz ElSayed Elgohary	20251701533	Cyber security
Eyad Waleed Ali Hassan	20251701506	Cyber security
Mostafa Ahmed Mostafa Mahmoud Ali Rabie	20251701536	Cyber security
Youssef Mohamed Ibrahim Arnous	20251701643	Artificial intelligence
Ahmed Ayman Fikry Abdelkhalek	20251701500	Cyber security
Abdelrahman Mohamed Talaat Arafa	20251701721	Software engineering
Mostafa Mahmoud Mohamed Alhadidi	20251701639	Artificial intelligence

2. Team Contributions

Team Member	Contributions / Modules Worked On
Eyad Waleed	Calculated salary, updated attendance, overall project structure, data validation, and bug fixing.
Mahmoud Elgohary	Admin-login and handle-admin menu.
Mostafa Rabie	Add employee and update employee functions.
Abdalrahman Mohammed	Delete employee and record attendance functions.
Ahmed Ayman	Employee login and view personal info functions.
Mostafa Al-Hadidi	View salary, view attendance, and delete attendance functions.
Youssef Arnous	Storage handling, overall project structure, data validation, and bug fixing.

3. System Architecture

Entity-Relationship Structure

- **Employee to Attendance (1:M):** One employee entity maps to multiple attendance records. They are linked via the employeeID attribute.
- **AdminAccount:** A standalone entity used strictly for administrative authentication and control.

System Architecture & Control Flow

- **Persistent Storage:** The layer maintaining the employees.txt, admins.txt, and attendance.txt files. Data is loaded sequentially into arrays upon startup and written back upon exiting.
- **Main Controller:** Prompts the user with the Main Menu. Based on the selected choice, the controller directs the flow to either the Admin Module or the Employee Module.
- **Admin Module:** Features Read/Write state privileges interacting directly with the global arrays (Runtime Memory).
- **Employee Module:** Features Read-Only state privileges, allowing users to securely view but not mutate global arrays.

4. System Overview

The Employee Payroll Management System is a lightweight, file-based C++ application that provides role-based access control for Administrators and Employees. It manages personal data, tracks working hours and monthly attendance, and automates salary calculations based on standard basic pay, overtime bonuses, and tax deductions. All data is persisted locally using text files.

5. Core Data Structures

Structure Name	Description
Employee	Stores all relevant personal, operational, and financial data for a single employee.
Attendance	Tracks the monthly attendance record for an employee, logging days present and absent.
AdminAccount	Stores credentials and details for administrative users.

6. User Roles & Access

6.1. Administrator Role

Administrators have full access to the system to manage personnel and payroll. The Admin menu includes the following capabilities:

- **Add/Update Employee:** Allows the admin to create a new employee profile or update an existing one.
- **Delete Employee:** Removes an employee from the system entirely.
- **Record/Update Attendance:** Logs the days present and days absent for a specific employee.
- **Delete Attendance:** Removes a logged attendance record.
- **Calculate Salary:** Triggers the system's financial calculation logic for a designated employee.
- **View All Employees Data:** Generates a comprehensive report displaying personal info, salary data, and attendance records.

6.2. Employee Role

Employees have restricted, read-only access limited strictly to their own data. They are allowed a maximum of 3 login attempts.

- **View Personal Information:** Displays Name, ID, Age, Phone, and Role.
- **View Salary Information:** Displays the latest calculated Basic Salary, Bonus, Overtime, Tax, and Net Salary.
- **View Attendance Record:** Shows all logged attendance data.

7. Salary Calculation Logic

The calculation handles the financial math for the payroll. The global constant TaxRate is set to 0.14 (14%).

1. **Overtime Calculation:** Overtime = TotalHoursWorked - WorkHoursPerMonth.
2. **Bonus Calculation:** Bonus = (Basic Salary / WorkHoursPerMonth) * Overtime.
3. **Tax Deduction:** Tax = Basic Salary * 0.14.
4. **Net Salary:** Net Salary = (Basic Salary + Bonus) - Tax.

8. Data Persistence

To ensure data is not lost when the program closes, the system utilizes text files.

- **employees.txt:** Stores all Employee structures.
- **admins.txt:** Stores AdminAccount structures.
- **attendance.txt:** Stores all attendance records.

9. Project Link

- https://github.com/Eyad-waleed/EmployeePayrollManagementSystem